



THREAT ADVISORY

**ATTACK
REPORT**

Unknown Iranian attackers leverage vulnerabilities to conduct ransom operations.

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Admiralty Code

A1

TA Number

TA2022204

Summary

The Iranian government-sponsored actors carry out malicious cyber activities against a wide range of people and entities in the United States, Australia, Canada, and the United Kingdom by using known vulnerabilities in Fortinet, Microsoft Exchange, Log4j, and VMware Horizon.

⚙️ CVE Table

CVE	NAME	PATCH
CVE-2021-44228	Apache Log4j Remote Code Execution Vulnerability	✓
CVE-2021-34473	Microsoft Exchange Server Remote Code Execution Vulnerability	✓
CVE-2018-13379	A path traversal vulnerability in FortiOS SSL VPN web portal	✓
CVE-2020-12812	FortiOS SSL VPN 2FA bypass	✓
CVE-2019-5591	Lack of LDAP server identity verification in default configuration	✓

CVE	NAME	PATCH
CVE-2021-45046	Apache Log4j2 Vulnerability	✓
CVE-2021-34470	Microsoft Exchange Server Elevation of Privilege Vulnerability	✓
CVE-2021-33766	Microsoft Exchange Information Disclosure Vulnerability	✓
CVE-2021-33768	Microsoft Exchange Server Elevation of Privilege Vulnerability	✓
CVE-2021-31206	Microsoft Exchange Server Remote Code Execution Vulnerability	✓
CVE-2021-45105	Apache Log4j denial of service	✓
CVE-2021-31196	Microsoft Exchange Server Remote Code Execution Vulnerability	✓
CVE-2021-34523	Microsoft Exchange Server Elevation of Privilege Vulnerability	✓
CVE-2021-31207	Microsoft Exchange Server Elevation of Privilege Vulnerability	✓



Potential MITRE ATT&CK TTPs

TA0001 Initial Access	T1190 Exploit Public-Facing Application	TA0002 Execution	T1053 Scheduled Task/Job
TA0042 Resource Development	T1588 Obtain Capabilities	T1588.001 Malware	T1588.002 Tool
T1053.005 Scheduled Task	T1486 Data Encrypted for Impact	TA0003 Persistence	T1136 Create Account
T1136.001 Local Account	T1136.002 Domain Account	TA0004 Privilege Escalation	TA0006 Credential Access
TA0009 Collection	TA0040 Impact	TA0010 Exfiltration	T1560 Archive Collected Data
T1560.001 Archive via Utility			

Technical Details

#1

Threat actors are exploiting known issues in Fortinet FortiOS(CVE-2018-13379, CVE-2020-12812, and CVE-2019-5591) and Microsoft Exchange server vulnerabilities (CVE-2021-34473) in conjunction with ProxyShell vulnerabilities (CVE-2021-34523 and CVE-2021-31207) to get initial access to the victim's network.

#2

In addition to these, the adversaries have exploited Log4j vulnerability (likely CVE-2021-44228, CVE2021-45046, and/or CVE-2021-45105) in a VMware Horizon application to gain access and laterally move across the compromised network.

#3

To further compromise the system, APT actors implemented tools to enable data exfiltration and extortion. The actors elevated privileges to gain administrator-level access to a network to maintain persistence. Finally, the ransom payment decryption keys were held in BitLocker through forced activation.

✂ Indicator of Compromise (IOC)

TYPE	VALUE
IPV4	54.39.78[.]148 95.217.193[.]86 104.168.117[.]149 107.173.231[.]114 144.76.186[.]88 148.251.71[.]182 172.245.26[.]118 185.141.212[.]131 198.12.65[.]175 198.144.189[.]74
Domains	newdesk[.]top symantecserver[.]co msupdate[.]us msupdate[.]top gupdate[.]us aptmirror[.]eu buylap[.]top winstore[.]us tcp443[.]org mssync[.]one upmirror[.]top

TYPE	VALUE
File Path	C:\Windows\Temp\wininet[.]xml C:\Windows\Temp\WinLogon[.]xml C:\Windows\wininet[.]bat C:\Windows\winlogon[.]bat C:\\ProgramData\Microsoft\CacheTask[.]bat C:\Windows\Temp\task_update[.]exe C:\Windows\dllhost[.]exe C:\Windows\svchost[.]exe C:\Windows\Temp\user[.]exe C:\Users\DefaultAccount\Desktop\New folder\setup[.]bat C:\Windows\Temp\ssasl[.]pmd C:\Windows\Temp\ssasl[.]zip C:\Users\DefaultAccount\Desktop\netscanold\netscanold[.]exe C:\Users\DefaultAccount\Desktop\scan[.]csv C:\Users\DefaultAccount\AppData\Local\Temp\lsass[.]dmp C:\Users\DefaultAccount\AppData\Local\Temp\lsass[.]zip
MD5	7fdc2d007ef0c1946f1f637b87f81590 Bd131ebfc44025a708575587afeebbf3 F0be699c8aafc41b25a8fc0974cc4582 68f58e442fba50b02130eedfc5fe4e5b 298d41f01009c6d6240bc2dc7b769205 0f8b592126cc2be0e9967d21c40806bc 9a3703f9c532ae2ec3025840fa449d4e 5b646edb1deb6396082b214a1d93691b Cacb64bdf648444e66c82f5ce61caf4b Ee8fd6c565254fe55a104e67cf33eaea 7ac4633bf064ebba9666581b776c548f 5f098b55f94f5a448ca28904a57c0e58 49c71178fa212012d710f11a0e6d1a30 2e1e17a443dc713f13f45a9646fc2179 d2f4647a3749d30a35d5a8faff41765e
SHA1	0f676bc786db3c44cac4d2d22070fb514b4cb64c E75bfc0dd779d9d8ac02798b090989c2f95850dc 226f0fbb80f7a061947c982ccf33ad65ac03280f 27102b416ef5df186bd8b35190c2a4cc4e2fbf37 524443dd226173d8ba458133b0a4084a172393ef 24ed561a1ddbced170acf1797723e5d3c51c2f5d 3a6431169073d61748829c31a9da29123dd61da8 763ca462b2e9821697e63aa48a1734b10d3765ee 3da45558d8098eb41ed7db5115af5a2c61c543af 8ece87086e8b5aba0d1cc4ec3804bf74e0b45bee 76dd6560782b13af3f44286483e157848efc0a4e 6ca62f4244994b5fbb8a46bdf62aa1c958cebbd 8b23b14d8ec4712734a5f6261aed40942 c9e0f68 6bae2d45bbd8c4b0a59ba08892692fe86 e596154

TYPE	VALUE
SHA256	559d4abe3a6f6c93fc9eae24672a49781af140c43d491a757c8e975507b4032e Bcc2e4d96e7418a85509382df6609ec9a53b3805effb7ddaed093bdaf949b6ea 668ec78916bab79e707dc99fdecfa10f3c87ee36d4dee6e3502d1f5663a428a0 D14d546070afda086a1c7166eaafd9347a15a32e6be6d5d029064bfa9ecde7de7 C1723fcad56a7f18562d14ff7a1f030191ad61cd4c44ea2b04ad57a7eb5e2837 12c6da07da24edba13650cd324b2ad04d0a0526bb4e853dee03c094075ff6d1a 17e95ecc7fedcf03c4a5e97317cfac166b337288562db0095ccd24243a93592f 724d54971c0bba8ff32aeb6044d3b3fd571b13a4c19cada015ea4bcab30cae26 1604e69d17c0f26182a3e3ff65694a49450aafd56a7e8b21697a932409dfd81e B04b97e7431925097b3ca4841b8941397b0b88796da512986327ff66426544ca 8aa3530540ba023fb29550643beb00c9c29f81780056e02c5a0d02a1797b9cd9 B8a472f219658a28556bab4d6d109fdf3433b5233a765084c70214c973becbbd 7b5fbbd90eab5bee6f3c25aa3c2762104e219f96501ad6a4463e25e6001eb00b

Patch Links

<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-34473>
<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-34523>
<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-31207>
<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-31196>
<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-31206>
<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-33768>
<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-33766>
<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-34470>
<https://www.vmware.com/security/advisories/VMSA-2021-0028.html>
<https://www.fortiguard.com/psirt/FG-IR-18-384>
<https://fortiguard.com/psirt/FG-IR-19-283>
<https://www.fortiguard.com/psirt/FG-IR-19-037>

References

<https://www.cisa.gov/uscert/sites/default/files/publications/aa22-257a-iranian-islamic-revolutionary-guard-corps-affiliated-cyber-actors.pdf>

What Next?

Book a free demo with **HivePro Uni5** to check your exposure to this advisory. HivePro Uni5 is a Continuous Threat Exposure Management Solution that proactively reduces an organization's attack surface before it gets exploited.



At Hive Pro we take a long hard look at your vulnerabilities so you can bolster your defenses and fine-tune your offensive cybersecurity tactics.

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